

Controlized Control (Inverse Dynamics controller)

$$B(q)\ddot{q} + \underbrace{\phi(q, \dot{q}) + g(q)}_{n(q, \dot{q})} = \tau \Rightarrow B\ddot{q} + n = \tau. \quad (1)$$

Let $\tau = \hat{n} + \hat{B}u$, then $B\ddot{q} + n = \hat{B}u + \hat{n}$. If the estimates $\hat{B}, \hat{n}$ are exact, then the dynamics become

$$\ddot{q} = u. \quad (2)$$

But the dynamics be governed according to the linear system

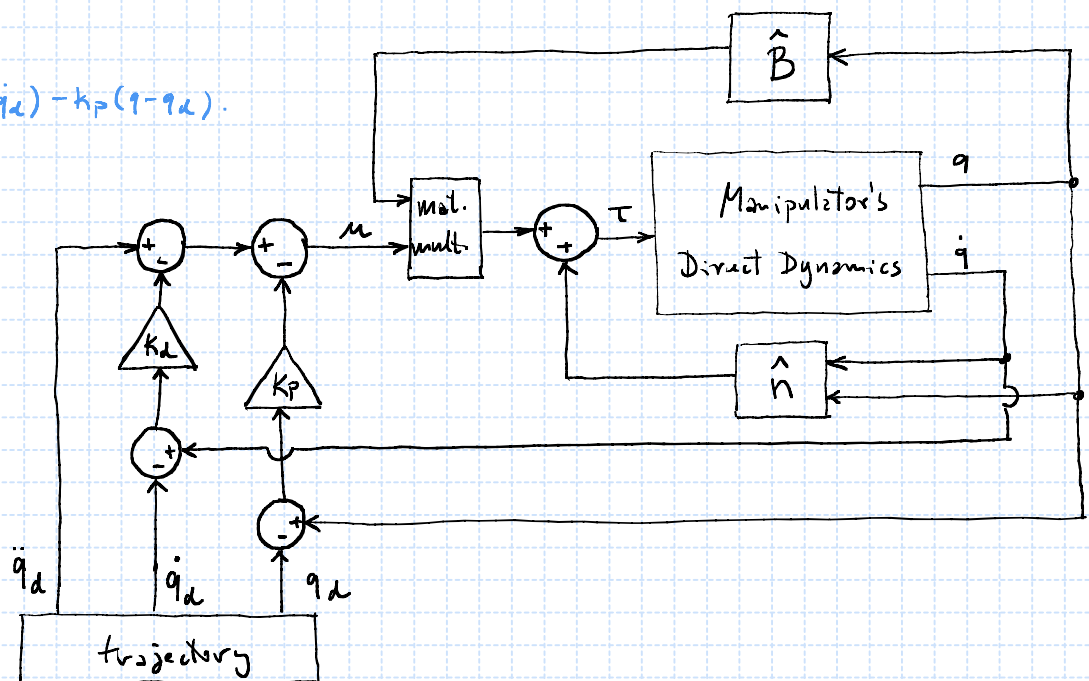
$$\ddot{e} + K_d \dot{e} + K_p e = 0, \text{ with } e \triangleq q - q_d. \quad (3)$$

Then, the input (u) must be

$$u = \ddot{q} = \ddot{q}_d - K_d(\dot{q} - \dot{q}_d) - K_p(q - q_d). \quad (4)$$

① (b) $\tau = \hat{n} + \hat{B}u$

$u = \ddot{q} = \ddot{q}_d - K_d(\dot{q} - \dot{q}_d) - K_p(q - q_d).$



① (a) $\omega_{n1} = 10 \text{ rad/s}, \omega_{n2} = 20 \text{ rad/s}, \zeta_1 = \zeta_2 = 1.$

$$K_d = \begin{pmatrix} K_{d1} & 0 \\ 0 & K_{d2} \end{pmatrix}, K_p = \begin{pmatrix} K_{p1} & 0 \\ 0 & K_{p2} \end{pmatrix}, e = \begin{pmatrix} q_1 - q_{1d} \\ q_2 - q_{2d} \end{pmatrix}.$$

$$\ddot{e} + K_d \dot{e} + K_p e = 0, \text{ with } e \triangleq q - q_d.$$

$$\ddot{e}_i + K_{d_i} \dot{e}_i + K_{p_i} e_i = 0$$

$$\text{Laplace Transform, } E_i(s) (s^2 + K_{d_i} s + K_{p_i}) = 0$$

$$\frac{0(s)}{I(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$K_{d_i} = 2\zeta_i \omega_{n_i}, \quad K_{p_i} = \omega_{n_i}^2$$

$$0(s) (s^2 + 2\zeta\omega_n s + \omega_n^2) = \underset{0}{\cancel{I(s)}} \omega_n^2$$

$$u = \ddot{q} = \ddot{q}_d - K_d (\dot{q} - \dot{q}_d) - K_p (q - q_d).$$

$$\begin{aligned} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} &= \begin{pmatrix} \ddot{q}_1 \\ \ddot{q}_2 \end{pmatrix} = \begin{pmatrix} \ddot{q}_{1d} - 2\zeta_1 \omega_{n_1} (\dot{q}_1 - \dot{q}_{1d}) - \omega_{n_1}^2 (q_1 - q_{1d}) \\ \ddot{q}_{2d} - 2\zeta_2 \omega_{n_2} (\dot{q}_2 - \dot{q}_{2d}) - \omega_{n_2}^2 (q_2 - q_{2d}) \end{pmatrix} \\ &= \begin{pmatrix} \ddot{q}_{1d} - 20 (\dot{q}_1 - \dot{q}_{1d}) - 100 (q_1 - q_{1d}) \\ \ddot{q}_{2d} - 40 (\dot{q}_2 - \dot{q}_{2d}) - 400 (q_2 - q_{2d}) \end{pmatrix} \end{aligned}$$

$$\omega_{n_1} = 10 \text{ rad/s}, \quad \omega_{n_2} = 20 \text{ rad/s}, \quad \zeta_1 = \zeta_2 = 1.$$

$$\tau = B u + n.$$

$$K_d = \begin{pmatrix} 2\zeta_1 \omega_{n_1} & 0 \\ 0 & 2\zeta_2 \omega_{n_2} \end{pmatrix}, \quad K_p = \begin{pmatrix} \omega_{n_1}^2 & 0 \\ 0 & \omega_{n_2}^2 \end{pmatrix}, \quad e = q - q_d$$